

MATH 350-2 Advanced Calculus

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Warm-up Challenge

Problem

Let $\vec{a} \in \mathbb{R}^n$.

Prove that the singleton set $\{\vec{a}\}$ is closed.

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- kinds of sets: open, closed, connected, component, compact
- kinds of functions: continuous, homeomorphism

Answers must be in terms of open sets

Definition

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- $\{(0, 1/2), (1/3, 2)\}$ is an open cover of the interval $(0, 1]$
- $\{(\frac{1}{n}, \frac{n+1}{n}) : n \in \mathbb{Z}_+\}$ is an open cover of $(0, 1]$

Lindelöf Covering Theorem

Theorem (Lindelöf Covering Theorem)

Let $A \subseteq \mathbb{R}^n$ be a set and suppose $\{U_i : i \in I\}$ is an open covering of A . Then there exists a countable subcover $\{U_j : j \in J\}$.

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A set $A \subseteq \mathbb{R}^n$ is called **compact** if every open cover of A has a *finite* subcover.

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Show that a singleton set $A = \{\vec{x}\}$ is compact.

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This means that $\vec{x} \in \bigcup_{i \in I} U_i$.

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It follows that $\{U_j\}$ is a subcover of A consisting of a single set.

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Thus every open cover has a finite subcover, making A compact!

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Suppose A is a compact set.

Consider the family of sets $\{U_i : i \in I\}$ where $I = \mathbb{Z}_+$ and $U_i = B(\vec{0}; i)$.



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Thus $\mathbb{R}^n = \bigcup_{i \in I} U_i$, and since $A \subseteq \mathbb{R}^n$, we see $A \subseteq \bigcup_{i \in I} U_i$.

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Solution

There is a finite subset $\{i_1, i_2, \dots, i_m\} \subseteq I$ with $A \subseteq \bigcup_{k=1}^m U_{i_k}$.

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This shows A is bounded.

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We will show $\vec{x}$ cannot be an adherent point of A and therefore A contains all of its adherent points.

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Consider the family of sets $\{U_i : i \in I\}$ where $I = \mathbb{Z}_+$ and $U_i = \mathbb{R}^n \setminus \overline{B}(\vec{x}; 1/i)$. □

Challenge

Problem

Prove that $\bigcup_{i \in I} U_i = \mathbb{R}^n \setminus \{\vec{x}\}$.

Challenge

Problem

Explain why $\{U_i : i \in I\}$ is an open cover of A .

Problem

Show that if $\{U_i : i \in I\}$ has a finite subcover, then $\vec{x}$ can't be an adherent point of $\vec{A}$.

Bolzano-Weierstrass Theorem

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Theorem (Bolzano-Weierstrass Theorem)

A bounded set $A \subseteq \mathbb{R}^n$ with infinitely many points must have an accumulation point.

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Suppose that $C_1, C_2, C_3, \dots$ are non-empty closed, bounded sets with $C_{i+1} \subseteq C_i$ for all $i \in I = \mathbb{Z}_+$. Then $\bigcap_{i \in I} C_i$ is also non-empty.

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Therefore the Bolzano-Weierstrass Theorem tells us it has an accumulation point $\vec{x}$.



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In fact, if we let $A_m = \{\vec{x}_i : i \in \mathbb{Z}_+, i \geq m\}$, then $\vec{x}$ is an accumulation point of A_m for all m .

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Moreover, $A_m \subseteq C_m$ so $\vec{x}$ is an accumulation point of C_m for all m .

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Since C_m is closed, it follows $\vec{x} \in C_m$, and thus $\vec{x} \in \bigcap_{i \in I} C_i$. In particular the intersection is non-empty! □

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Then by Lindelöf's Theorem, we can take a countable subcover.

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Therefore without loss of generality, assume $I = \mathbb{Z}_+$.

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$$C_m = A \cap \left(\mathbb{R}^n \setminus \bigcup_{i=1}^m U_i \right).$$



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Notice that $C_{m+1} \subseteq C_m$ for all i .

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Notice that $C_{m+1} \subseteq C_m$ for all i .

Since they are non-empty, closed and bounded, the Cantor Intersection Theorem says $\bigcap_{m=1}^{\infty} C_m$ is non-empty.

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To prove this must happen, we assume otherwise.

Notice that $C_{m+1} \subseteq C_m$ for all i .

Since they are non-empty, closed and bounded, the Cantor Intersection Theorem says $\bigcap_{m=1}^{\infty} C_m$ is non-empty.

Let $\vec{x} \in \bigcap_{m=1}^{\infty} C_m$.

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A subset $A \subseteq \mathbb{R}^n$ is compact if and only if it is closed and bounded.

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Since $\vec{x} \in A \subseteq \bigcup_{i \in I} U_i$, we have that $\vec{x} \in U_k$ for some $k \in I$, meaning $\vec{x} \notin C_k$.

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This is a contradiction. □